

The rapidity term

Is $2b$ the only one — and does it build the evolution kernel?

$$K_{ab} \equiv \frac{a-b}{(a-b)^2} \text{ the Weizsäcker–Williams kernel; } A = (x-z)^2, B = (x-w)^2, A' = (x'-z)^2, B' = (x'-w')^2$$

Your rewriting is exact, and within this row the answer is yes — the blue piece is the only source of $\log \frac{1}{\Lambda}$; even the other two pieces of $2b$ are Λ -finite (§2). **But it is not the only term in the cross section, and the proof is inside this very expression.** Your prefactor collapses to $\frac{1}{(x-w)^2(x'-w')^2}$ and your four-term bracket factorises (§3), leaving the z -kernel

$$\mathcal{V}_z = K_{zw} - \frac{1}{2}K_{zx},$$

which is **not** the dipole kernel $K_{zw} - K_{zx}$. That factor of $\frac{1}{2}$ is exactly why this row has a $|z| \rightarrow \infty$ logarithm: the half-weighted kernel squared has a $\frac{1}{4|z|^2}$ tail, while the full one falls as $|z|^{-4}$ (§4). The row's IR divergence and the incompleteness of its evolution kernel are **the same fact**, and they close together when the partner rows are added.

1. The blue step

$$\log \frac{\vee - k^+}{\Lambda} = \log \frac{\vee}{\Lambda} + \log \left(1 - \frac{k^+}{\vee}\right)$$

exact (symbolic difference 0). The second term is finite and $\rightarrow 0$ for $\vee \gg k^+$, so the rapidity logarithm is $\log \frac{\vee}{\Lambda}$ and the rest is an ordinary kinematic correction. Splitting it off the way you have is the right thing to do.

2. Within this row, yes — and only this piece

part	$\Lambda = 10^{-5}$	$\Lambda = 10^{-9}$	$\partial/\partial \log \frac{1}{\Lambda}$	predicted
$2a$	0.024054701	0.024055355	$7 \cdot 10^{-8}$	0
$2b$	0.621193354	1.223661304	0.065412127	0.065412525
$2c$	0.125796664	0.125796664	0	0
$2d$	0.015838295	0.015838949	$7 \cdot 10^{-8}$	0
$2e$	0.008216406	0.008216406	0	0
$2f$	0.000359660	0.000359660	0	0

The predicted residue is $\frac{1}{k^+(x-w)^2(x'-w')^2}$. And inside $2b$ itself, its second and third pieces are Λ -stable ($-2.129081618 \rightarrow -2.129086767$ and $1.093478078 \rightarrow 1.093479556$ over four decades of Λ) — because $\log \frac{(\vee - k^+)A + k^+B}{\Lambda A + k^+B} \rightarrow \log \frac{(\vee - k^+)A + k^+B}{k^+B}$, which is finite. So the blue piece alone.

3. Two simplifications you are leaving on the table

3.1 The prefactor is just $\frac{1}{(x-w)^2(x'-w')^2}$

$$\frac{1}{B'AB - B^2A'} \times \left[A - \frac{BA'}{B'} \right] = \frac{1}{BB'} \quad \text{identically.}$$

(Symbolic; numerically 0.065412525179 both ways.) The $\Delta = AB' - A'B$ cancels between the two factors — which also means **this piece has no removable pole at all**, unlike every other piece in the six parts. It is the one term you can integrate numerically without any special care.

3.2 The four-term bracket factorises

$$\left[\cdots \text{your four terms} \cdots \right] = [(x-w) \cdot (x'-w')] [\mathcal{V}_z \cdot \mathcal{V}'_z],$$

$$\mathcal{V}_z = \frac{z-w}{(z-w)^2} + \frac{x-z}{2(x-z)^2} = K_{zw} - \frac{1}{2}K_{zx}, \quad \mathcal{V}'_z = K_{zw'} - \frac{1}{2}K_{zx'}.$$

(Checked: 0.036521945575 both ways.) Putting the two together, your whole rapidity term is

$$\frac{1}{(2\pi)^3} \frac{g^4}{8\pi^5} \frac{1}{k^+} \log \frac{\sqrt{\Lambda}}{\Lambda} \int_{z,w,w',x,x'} e^{-ik(w'-w)} \underbrace{\frac{(x-w) \cdot (x'-w')}{(x-w)^2(x'-w')^2}}_{\text{LO structure}} \underbrace{\left[K_{zw} - \frac{1}{2}K_{zx} \right] \cdot \left[K_{zw'} - \frac{1}{2}K_{zx'} \right]}_{\text{Z-gluon kernel}} \times [\text{colour}]$$

which is the shape a rapidity term must have: the LO single-gluon-production structure times a kernel for the emitted soft gluon, times the rapidity span. Worth writing it this way — it is much easier to compare against the JIMWLK kernel.

4. No, it is not the whole evolution kernel — and the $\frac{1}{2}$ proves it

The BK/JIMWLK dipole kernel is $K_{zw} - K_{zx}$: the **difference of two full** WW kernels, one to each end of the dipole. What you have is $K_{zw} - \frac{1}{2}K_{zx}$. The missing half has to come from somewhere, and the place it shows up is the large- $|z|$ behaviour:

$ z $	yours: $(K - \frac{1}{2}K) \cdot (K - \frac{1}{2}K) \times z ^2$	full BK: $(K - K) \cdot (K - K) \times z ^2$
10^2	0.250238667	0.000261103
10^3	0.250002390	0.000002612
10^4	0.250000024	0.000000026
10^5	0.250000000	0.000000000

Yours tends to $\frac{1}{4}$ — a flat $\frac{1}{4|z|^2}$ tail, hence $\int \frac{d^2z}{|z|^2}$, a logarithm. The full kernel tends to zero like $|z|^{-2}$, i.e. the product falls as $|z|^{-4}$: **no large- $|z|$ log at all**. The reason is the dipole structure itself — at large $|z|$, K_{zw} and K_{zx} both approach $\hat{z}/|z|$ and cancel to leading order, leaving only the $O(|z|^{-2})$ difference. Colour neutrality in position space.

So the row's IR divergence and its incomplete kernel are one and the same fact. Multiplying this term by $\frac{K_{ww'}}{k^+}$ and the $\frac{1}{4}$ reproduces exactly the $|z| \rightarrow \infty$ coefficient measured earlier, $\frac{K_{ww'}}{4k^+} \log \frac{\sqrt{-k^+}}{\Lambda}$ — the same number, arrived at two ways. When the partner rows supply the other halves of K_{zx} and $K_{zx'}$, the kernel becomes $K_{zw} - K_{zx}$ and the z integral converges at the same moment. You should not try to regulate that log on its own; it is telling you the sum is incomplete.

5. What still has to contribute

To finish the kernel and close the z integral you need, at minimum:

- the $(1\rho) \times (2\rho)$ cross terms, your N_1 — whose $|z| \rightarrow \infty$ plateau I measured earlier as exactly $-2 \times$ this row's;
- the $(2\rho) \times (2\rho)$ piece of $|\mathfrak{B}_2|^2$, still missing from your expression, coefficient $\frac{1}{(2\pi)^3} \frac{g^4}{16\pi^4} \frac{1}{p^+k^+}$ — note its $\frac{1}{p^+}$, which is a rapidity log of its own;
- the $\mathfrak{B}_3$ rows, which also carry $\frac{1}{p^+}$;
- the virtual rows — these supply the opposite-sign half of the real-virtual cancellation and are what turn the kernel into a difference in the first place.

Only once all of those are in should you read off the coefficient of $\log \frac{\sqrt{\Lambda}}{\Lambda}$ and match it to the JIMWLK Hamiltonian acting on the LO cross section.

Checked by `src/row21_rap.py`; output in `RESULTS_row21.txt`.